Elementary Analysis

Section 29: The Mean Value Theorem

Exercises 29.1–29.18, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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29 The Mean Value Theorem

The Mean Value Theorem (MVT) connects an average slope to an actual derivative. If f is continuous on $[a, b]$ and differentiable on (a, b), with $a < b$, there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

The theorem asserts existence, not a prescribed choice of c. Both hypotheses matter. Ross uses "increasing" to mean nondecreasing; "strictly increasing" requires strict inequalities. A derivative sign on an entire interval controls monotonicity there; a sign at one point does not have the same force.

29.1 Exercise 29.1: Checking the MVT conclusion in six examples

What we must show. Compute the secant slope and determine whether an interior derivative equals it.

(a) The square on $[-1, 2]$.

The slope is $(4 - 1)/3 = 1$. Since $f'(x) = 2x$, take $c = 1/2 \in (-1, 2)$. Polynomial continuity and differentiability verify all MVT hypotheses.

(b) Sine on $[0, \pi]$.

The secant slope is 0, and $f'(\pi/2) = \cos(\pi/2) = 0$. Sine satisfies all hypotheses.

(c) Absolute value on $[-1, 2]$.

The secant slope is $(2 - 1)/3 = 1/3$. At nonzero interior points the derivative is either -1 or 1; at 0 it is undefined. Thus no derivative equals $1/3$. The failed hypothesis is differentiability throughout the open interval, although continuity holds.

(d) The reciprocal on $[-1, 1]$.

The endpoint expression gives $(1 - (-1))/2 = 1$. Wherever it exists, the derivative is $-1/x^2 < 0$, so it cannot equal 1. The function is not even defined at 0; in particular it is neither continuous on the whole closed interval nor differentiable on the whole open interval.

(e) The reciprocal on $[1, 3]$.

Now both hypotheses hold. The slope is $(1/3 - 1)/2 = -1/3$, and $-1/c^2 = -1/3$ is satisfied by $c = \sqrt{3} \in (1, 3)$.

(f) The sign function on $[-2, 2]$.

The secant slope is $(1 - (-1))/4 = 1/2$. At every nonzero point its derivative is 0; at 0 it has a jump and no derivative. Thus the conclusion fails. Continuity on the closed interval and differentiability throughout the open interval both fail.

Conclusion. The conclusion holds in (a), (b), and (e), at the exhibited points; it fails in (c), (d), and (f).

29.2 Exercise 29.2: Cosine is Lipschitz with constant one

What we must show. Prove $|\cos x - \cos y| \le |x - y|$ for all real x, y.

Handle equal points and order unequal ones.

If $x = y$, both sides vanish. Otherwise interchange their names if needed so $x < y$. Cosine is continuous on $[x, y]$ and differentiable on (x, y). The MVT gives $c \in (x, y)$ with

$$\cos y - \cos x = -\sin c\,(y - x).$$

Taking absolute values and using $|\sin c| \le 1$ proves the inequality. The absolute-value form is unchanged when the original points are interchanged.

29.3 Exercise 29.3: Two guaranteed derivative values

What we must show. Use $f(0) = 0$, $f(1) = f(2) = 1$ to locate derivative values $1/2$ and $1/7$.

(a) Use the endpoints 0 and 2.

Differentiability on $\mathbb{R}$ implies continuity there, so the MVT applies on $[0, 2]$. It supplies $c \in (0, 2)$ with

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{1}{2}.$$

(b) First find slopes above and below 1/7.

The MVT on $[0, 1]$ gives $u \in (0, 1)$ with $f'(u) = 1$. On $[1, 2]$ it gives $v \in (1, 2)$ with $f'(v) = 0$. Derivatives have the intermediate value property (Darboux's theorem, Theorem 29.8), even when they are not continuous. Therefore some $w \in (u, v) \subset (0, 2)$ satisfies $f'(w) = 1/7$.

An elementary way to implement the same argument.

Let $H(x) = f(x) - x/7$ on $[u, v]$. Then $H'(u) = 6/7 > 0$ and $H'(v) = -1/7 < 0$. By the derivative definition, H takes values larger than $H(u)$ just to the right of u, and values larger than $H(v)$ just to the left of v. Its maximum on the compact interval therefore cannot occur at either endpoint. At an interior maximum w, Fermat's theorem gives $H'(w) = 0$, or $f'(w) = 1/7$. This explains why continuity of f' is not needed.

29.4 Exercise 29.4: Introducing an exponential integrating factor

What we must show. Find $c \in (a, b)$ with $f'(c) + f(c)g'(c) = 0$ when $f(a) = f(b) = 0$.

Plan. The derivative of fe^g contains exactly the desired expression, multiplied by a positive factor.

Verify Rolle's theorem for the auxiliary function.

Define $H(x) = f(x)e^{g(x)}$. The differentiability assumptions on the open interval I ensure that H is continuous on $[a, b] \subset I$ and differentiable on (a, b). Also $H(a) = H(b) = 0$. Rolle's theorem gives $c \in (a, b)$ with $H'(c) = 0$.

Differentiate and divide by a nonzero factor.

By the product and chain rules,

$$0 = H'(c) = e^{g(c)}[f'(c) + f(c)g'(c)].$$

Because $e^{g(c)} > 0$, division yields the required equation.

29.5 Exercise 29.5: A quadratic bound forces constancy

What we must show. Show a function satisfying $|f(x) - f(y)| \leq (x-y)^2$ for all real x, y is constant.

Prove differentiability rather than assume it.

Fix $a \in \mathbb{R}$. For $h \neq 0$, the assumed estimate gives

$$\left|\frac{f(a+h) - f(a)}{h}\right| \leq \frac{h^2}{|h|} = |h| \longrightarrow 0.$$

Thus $f'(a)$ exists and equals 0. Since a was arbitrary, $f' = 0$ everywhere.

Use the MVT between any two points.

For $x < y$, the now-established differentiability gives the MVT hypotheses on $[x, y]$. Hence $f(y) - f(x) = f'(c)(y - x) = 0$ for some c between them. Every pair has equal values, so f is constant.

29.6 Exercise 29.6: The straight line in the MVT proof

What we must show. Write the line through $(a, f(a))$ and $(b, f(b))$ explicitly.

Use point–slope form.

The required function is

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Indeed $L(a) = f(a)$ and $L(b) = f(a) + f(b) - f(a) = f(b)$. Its derivative is the constant secant slope $(f(b) - f(a))/(b - a)$. Subtracting this line from f gives a function with equal endpoint values, which is precisely why Rolle's theorem can be used in the MVT proof.

29.7 Exercise 29.7: Vanishing higher derivatives imply polynomial form

What we must show. Determine f when its second or third derivative vanishes on an interval.

(a) If $f'' = 0$, first make f' constant.

The zero-derivative corollary to the MVT, applied to f', gives a constant A with $f'(x) = A$ on I. Now $[f(x) - Ax]' = 0$ on I, so the same corollary gives another constant B with $f(x) - Ax = B$. Thus $f(x) = Ax + B$.

(b) If $f''' = 0$, repeat one level higher.

Apply the corollary to f'' to obtain $f''(x) = C$. Then

$$[f'(x) - Cx]' = 0,$$

so $f'(x) = Cx + D$. Finally,

$$\left[f(x) - \frac{C}{2}x^2 - Dx\right]' = 0,$$

which gives $f(x) = \frac{C}{2}x^2 + Dx + E$. Conversely, every polynomial of this form has third derivative zero, so these are exactly the possibilities.

Conclusion. The interval assumption allows the MVT to connect every pair of points. On disconnected domains the constants could differ between components.

29.8 Exercise 29.8: Derivative signs and the three remaining monotonicity rules

What we must show. Prove Corollary 29.7(ii)–(iv).

Use one MVT equation for all three cases.

Choose arbitrary $x_1 < x_2$ in the interval. The MVT gives $c \in (x_1, x_2)$ with

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

The factor $x_2 - x_1$ is strictly positive, so it preserves the sign of $f'(c)$.

(ii) Strictly negative derivative.

If $f' < 0$ everywhere, the right side is negative. Thus $f(x_2) < f(x_1)$ for every ordered pair, which is strict decrease.

(iii) Nonnegative derivative.

If $f' \geq 0$ everywhere, then $f(x_2) - f(x_1) \geq 0$, so $f(x_1) \leq f(x_2)$. This is increase in Ross's nonstrict sense.

(iv) Nonpositive derivative.

If $f' \leq 0$ everywhere, the equation gives $f(x_2) \leq f(x_1)$, which is decrease. Equality is permitted in the last two cases.

29.9 Exercise 29.9: The inequality $ex \leq e^x$

What we must show. Show the product $e \cdot x$ never exceeds e^x.

Subtract and locate the minimum.

Let $F(x) = e^x - ex$. Then $F'(x) = e^x - e$. Because the exponential is strictly increasing, this derivative is negative for $x < 1$, zero at 1, and positive for $x > 1$. The monotonicity corollaries imply F decreases toward 1 and increases afterward. Therefore its global minimum is

$$F(1) = e - e = 0.$$

Hence $e^x - ex \geq 0$ for every real x.

Conclusion. The inequality holds for every real x, with equality exactly when $x = 1$. Here ex denotes multiplication, not a second copy of e^x.

29.10 Exercise 29.10: A positive derivative at one point does not imply local increase

What we must show. Study $f(x) = x^2 \sin(1/x) + x/2$ for $x \neq 0$, with $f(0) = 0$.

(a) Differentiate at zero.

The difference quotient is $x \sin(1/x) + 1/2$, whose first term tends to zero by $|x \sin(1/x)| \leq |x|$. Thus $f'(0) = 1/2 > 0$.

(b) Find negative derivatives arbitrarily close to zero.

For $x \neq 0$, Exercise 28.4 and the sum rule give

$$f'(x) = 2x \sin(1/x) - \cos(1/x) + \frac{1}{2}.$$

At $x_n = 1/(2\pi n)$ this equals $-1/2$. Every open interval containing zero contains such an x_n as an interior point. An increasing function that is differentiable at an interior point has nonnegative derivative there: its difference quotients are nonnegative for increments of either sign. Thus the negative derivative at x_n rules out increase on that interval.

(c) Compare the hypotheses carefully.

Corollary 29.7(i) requires $f'(x) > 0$ at *every* point of the interval. This example gives positivity only at zero and negative values arbitrarily nearby, so there is no contradiction.

29.11 Exercise 29.11: Sine lies below its argument on the positive half-line

What we must show. Prove $\sin x \leq x$ for $x \geq 0$.

Apply a derivative sign to the difference.

Let $F(x) = x - \sin x$. Then $F'(x) = 1 - \cos x \geq 0$ on $\mathbb{R}$, so F is increasing. For $x \geq 0$ this implies $F(x) \geq F(0) = 0$, which rearranges to the required inequality. At $x = 0$ equality is immediate.

29.12 Exercise 29.12: Three related trigonometric inequalities

What we must show. Establish the tangent inequality, monotonicity of $x/\sin x$, and the sine lower bound on a quarter period.

(a) Compare tangent with the identity.

Let $F(x) = \tan x - x$ on $[0, \pi/2)$. It is continuous at zero, $F(0) = 0$, and for $0 < x < \pi/2$,

$$F'(x) = \sec^2 x - 1 = \tan^2 x > 0.$$

Applying the MVT on $[0, x]$ gives $F(x) > F(0) = 0$. Thus $x < \tan x$.

(b) Differentiate the quotient with a positive denominator.

On $(0, \pi/2)$,

$$\left(\frac{x}{\sin x}\right)' = \frac{\sin x - x \cos x}{\sin^2 x}.$$

Both $\cos x$ and $\sin^2 x$ are positive. Part (a), multiplied by $\cos x > 0$, gives $x \cos x < \sin x$. Therefore the numerator is positive, and the quotient is strictly increasing.

(c) Compare with the endpoint value.

The quotient extends continuously to $x = \pi/2$ with value $\pi/2$. Its increase on the interior yields $x/\sin x \leq \pi/2$ for $0 < x \leq \pi/2$, for instance by taking an increasing sequence of comparison points tending to the endpoint. Multiplying by the positive $\sin x$ gives $x \leq (\pi/2)\sin x$. At $x = 0$ both sides vanish.

Conclusion. The last inequality is equivalently $\sin x \geq 2x/\pi$ on $[0, \pi/2]$.

29.13 Exercise 29.13: Comparing functions by comparing derivatives

What we must show. Use equal initial values and $f' \leq g'$ to show $f(x) \leq g(x)$ for $x \geq 0$.

Build a single function with a known sign of derivative.

Set $H = g - f$. Then $H(0) = 0$ and $H' = g' - f' \geq 0$. The monotonicity corollary makes H increasing. Hence for $x \geq 0$, $H(x) \geq H(0) = 0$, or $f(x) \leq g(x)$.

Conclusion. Equal initial values anchor the comparison. Without them, derivative inequalities alone do not fix which graph starts higher.

29.14 Exercise 29.14: Bounding a function between two straight lines

What we must show. Given $f(0) = 0$ and $1 \leq f' \leq 2$, prove $x \leq f(x) \leq 2x$ for $x \geq 0$.

Use the MVT on the interval from zero to the chosen point.

At $x = 0$ the assertion is immediate. For $x > 0$, differentiability ensures the hypotheses on $[0, x]$, so for some $c \in (0, x)$,

$$f(x) - f(0) = f'(c)x.$$

Since $x > 0$, multiplying $1 \leq f'(c) \leq 2$ by x preserves order. Substituting $f(0) = 0$ gives $x \leq f(x) \leq 2x$.

29.15 Exercise 29.15: The power rule for rational exponents

What we must show. Prove the derivative formula for $h(x) = x^{m/n}$ at every permitted nonzero real input, including negative inputs when n is odd.

Differentiate the real root by the inverse-function theorem.

Let $u(x) = x^{1/n}$ on its real domain. Fix a permitted $a \neq 0$ and write $t = u(a)$, so $t^n = a$ and $t \neq 0$. The function $v(t) = t^n$, restricted to the positive half-line if n is even and considered on $\mathbb{R}$ if n is odd, is one-to-one and continuous. Near this nonzero t it is differentiable with $v'(t) = nt^{n-1} \neq 0$. The inverse derivative theorem gives

$$u'(a) = \frac{1}{nt^{n-1}}.$$

Compose with an integer power.

The function $h = u^m$ is differentiable near a; negative m is allowed because u is nonzero there. The integer power rule and chain rule give

$$h'(a) = mt^{m-1}\frac{1}{nt^{n-1}} = \frac{m}{n}t^{m-n} = \frac{m}{n}a^{m/n-1}.$$

The final equality uses the real-root definition of the rational powers. It remains valid for negative a when n is odd, without invoking an undefined logarithm of a negative number.

Conclusion. $h'(a) = ra^{r-1}$ everywhere requested. The exclusion of zero avoids both inverse-derivative failure and negative-power domain issues.

29.16 Exercise 29.16: The derivative of arctangent

What we must show. Apply the inverse derivative theorem to tangent on $(-\pi/2, \pi/2)$.

Check invertibility and derivative hypotheses.

On this interval, $f(x) = \tan x$ is continuous and strictly increasing, has range $\mathbb{R}$, and has derivative $\sec^2 x > 0$. It therefore has the inverse $g = \arctan : \mathbb{R} \to (-\pi/2, \pi/2)$, and Theorem 29.9 applies at every inverse pair $y = \tan x$.

Express the derivative using the new input.

The theorem gives $g'(y) = 1/\sec^2 x$. Since $\sec^2 x = 1 + \tan^2 x = 1 + y^2$, this becomes

$$\frac{d}{dy}\arctan y = \frac{1}{1+y^2} \qquad (y \in \mathbb{R}).$$

29.17 Exercise 29.17: Matching values and slopes at a join

What we must show. Prove the criterion for differentiability of h, formed from f to the left of a and g at and to its right.

Necessity of matching values.

Suppose h is differentiable at a. It is then continuous there. By continuity of f,

$$f(a) = \lim_{x\to a^-} f(x) = \lim_{x\to a^-} h(x) = h(a) = g(a).$$

Thus the two branches meet at the same height.

Necessity of matching slopes.

Using $h(a) = f(a) = g(a)$, the left difference quotient for h has limit $f'(a)$, while the right one has limit $g'(a)$. Both must equal $h'(a)$, so $f'(a) = g'(a)$.

Sufficiency.

Conversely, suppose both equalities hold. For $x < a$ the difference quotient of h at a is $(f(x) - f(a))/(x - a)$; for $x > a$ it is $(g(x) - g(a))/(x - a)$. Their limits are the equal numbers $f'(a)$ and $g'(a)$. The two-sided limit therefore exists, and $h'(a)$ is their common value.

Conclusion. Matching heights prevents a jump; matching slopes prevents a corner. Both conditions are necessary and together are sufficient.

29.18 Exercise 29.18: Iterating a contraction on the real line

What we must show. Prove convergence of $s_{n+1} = f(s_n)$ and obtain a fixed point when $a = \sup_{x\in\mathbb{R}} |f'(x)| < 1$.

(a), first stage: convert the derivative bound to a distance bound.

Since $|f'| \geq 0$, we have $0 \leq a < 1$. The MVT gives, for any $u, v \in \mathbb{R}$,

$$|f(u) - f(v)| \leq a|u - v|.$$

For equal inputs the inequality is automatic. Applying it to successive iterates gives

$$|s_{n+1} - s_n| \leq a|s_n - s_{n-1}| \leq \cdots \leq a^n|s_1 - s_0|.$$

(a), second stage: sum the small successive gaps.

Assume first $0 < a < 1$. For integers $m > n$, the triangle inequality and a geometric series give

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| \leq |s_1 - s_0| \sum_{k=n}^{m-1} a^k \leq \frac{|s_1 - s_0|a^n}{1 - a}.$$

This bound tends to zero independently of $m > n$. For any $\varepsilon > 0$, choose N to make the last expression less than ε for $n \geq N$; this is exactly the Cauchy criterion. Completeness of $\mathbb{R}$ gives a limit s. If $a = 0$, the distance bound says f is constant, so all iterates from s_1 onward are equal and convergence is immediate.

(b) Pass to the limit in the iteration.

Differentiability implies continuity, and therefore

$$f(s) = f\left(\lim_n s_n\right) = \lim_n f(s_n) = \lim_n s_{n+1} = s.$$

Thus s is a fixed point. In fact it is unique: if $t = f(t)$, then $|s - t| = |f(s) - f(t)| \leq a|s - t|$; since $1 - a > 0$, this forces $s = t$.

Conclusion. Every starting value produces a convergent sequence with the same fixed-point limit. The geometric bound explicitly explains why the iteration settles down.